

CHARAN YAKASIRI

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PROFESSIONAL SUMMARY

Aspiring *FPGA and Low-Level Systems Engineer* focused on building efficient digital systems and growing across RTL, system-level architecture, and chip design.

EDUCATION

B.E – Electronics and Communication Engineering
Saveetha Engineering College, Chennai

JUN 2023 - MAY 2027
GPA: 8.0

Higher Secondary Education – PCM (Class XII)
Terapanth Jain Vidyalaya Mat. Hr. Sec. School, Chennai

JUN 2021 - MAR 2023
GPA: 8.5

PROFESSIONAL EXPERIENCE

Combat Vehicle Research and Development Establishment
Intern (EMI/EMC)

Chennai, India
JAN 2025

- Executed 8+ EMI/EMC tests using spectrum analyzers; maintained 20+ compliant test records for automotive and defense systems.
- Evaluated noise immunity across multiple scenarios, applying shielding techniques to reduce interference by 20–30%.

SKILLS

Core Skills: RTL Design, FPGA Synthesis & Implementation, FSM Design, IP Integration, Low-Level System Design

Languages: Verilog HDL, SystemVerilog, C (Bare Metal), Python, TCL

Technologies: FPGA, RISC-V ISA, Hardware-Software co-design, Timing Analysis, RTOS

Tools: Xilinx Vivado, Intel Quartus, ModelSim, Yosys, nextpnr, OpenROAD, QNX Momentics IDE, Cadence Virtuoso

Boards: PYNQ-Z2, Spartan-7, VSDSquadron FM, Vicharak Shrike Lite

Other: Linux, Git, Shell Scripting

PROJECTS

RTL-Based CNN Accelerator on FPGA (PYNQ-Z2) | [GitHub](#)

- Designed *synthesizable RTL convolution IP* in Verilog for PYNQ-Z2, achieving *~0.89 ms latency and ~1121 FPS throughput* under test conditions.
- Led FPGA implementation and Hardware-Software co-design, integrating AXI (Lite/Stream) IP and enabling *low-latency data transfer validated over 1000+ executions*
- Developed *Python-based preprocessing & postprocessing pipeline* and performed *system validation across 1000+ runs*, analyzing latency, throughput, and functionality.

Custom IP Integration on RISC-V (FPGA) | [GitHub](#)

- Optimized *GPIO & PWM IPs* in Verilog, achieving less than *5% LUT utilization* and *sub-10ns latency* via memory-mapped RISC-V integration.
- Developed *2KB bare-metal C firmware* to validate hardware, achieving *100% test coverage* across PWM frequencies using Yosys / IceStorm on VSDSquadron FM.

CERTIFICATIONS & ACHIEVEMENTS

- Samsung ISWDP Fellow (Cohort 4)** – Gained exposure to semiconductor manufacturing and Synopsys TCAD tools
- FPGA Design Program – VSD** – Built and integrated custom IPs (GPIO, PWM) with RISC-V on VSDSquadron FM
- RTOS using QNX – Pi Square Technologies** – Studied QNX microkernel architecture using Momentics IDE.
- Contributed to NSS initiatives by *teaching and improving rural school infrastructure*, enhancing learning conditions
- Organized and hosted *school and college events*, managing execution and strengthening leadership and communication skills